



WSQ · BUSINESS INNOVATION

Business Innovation with AI Agents

MODEL + HARNESS → GOAL-DIRECTED BUSINESS ACTION

TGS-2020503395

**Opportunities · feasibility · Hermes harness
Multi-agent campaigns · security · human oversight**

Tertiary Infotech Academy Pte Ltd · v23.0 · 11 September 2026



Welcome & Housekeeping

Set up the conditions for focused, safe and evidence-led learning.

ATTEND

PARTICIPATE

PROTECT DATA

VERIFY

Digital Attendance (Mandatory)

AM

Scan and submit at the start of each training day.

PM

Complete the afternoon record before leaving.

EXAM

A separate attendance record is required for assessment.

75%

Minimum attendance is required for assessment and funding eligibility.

About the Trainer



General Trainer

WSQ Facilitator · Technology and
Business Innovation

ROLE

Facilitates discussion, practice and assessment readiness

EXPERTISE

Business process improvement, AI adoption and
responsible innovation

FOCUS

Transfer concepts into safe workplace experiments

APPROACH

Explain → demonstrate → practise → verify

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech
Academy

ROLE

Founder and Principal Trainer

EXPERTISE

AI, automation, data and software systems

FOCUS

Practical workplace application and governance

APPROACH

Architecture → runtime trace → lab evidence

Let's know each other

1

Your role

What work do you want an agent to improve?

2

Your experience

Which AI or automation tools have you used?

3

Your goal

What safe, observable outcome do you want by the end of today?

Ground rules

1

Participate

Ask questions and share evidence.

2

Protect data

Use fictional data; never expose live secrets.

3

Verify

Test outputs before calling work complete.

4

Respect

One conversation at a time; challenge ideas, not people.

One-day Learning Journey

15 min

Welcome, AM digital attendance and assessment briefing

45 min

Install Hermes and engineer a task context

45 min

Prioritise ten industry use cases

65 min

Tools, MCP, memory and skills

45 min

Subagents and marketing campaign

45 min

Lab 8 and integrated course review

30 min**Oral Questioning (OQ)****50 min**

AI Engineering Evolution and Agent Architecture

40 min

Business Opportunities and Feasibility

45 min

Hermes Agent Harness in Practice

40 min

Multi-Agent Campaign Orchestration

15 min

Security Governance and Implementation

30 min**Written Assessment (SAQ), open book**

What You Will Be Able to Do

LO1

Compare the 2023 prompt, 2024 context, 2025 harness and 2026 agent engineering paradigms and evaluate their impact on business models.

LO2

Identify and prioritise realistic AI-agent opportunities using process evidence, value measures and human accountability.

LO3

Digitalise a bounded workflow with Hermes Agent tools, context, memory, skills and specialist subagents.

LO4

Conduct feasibility and cost-benefit analysis for an AI-agent pilot using measurable assumptions and risk-adjusted controls.

LO5

Implement and evaluate a governed multi-agent marketing or video campaign with security, data governance and human release approval.

Final Assessment

**30
min**

Written Assessment (SAQ)

Five open-ended questions mapped to A1–A5 and the taught business-innovation process.

Competent / Not Yet Competent • Assessor records evidence

**30
min**

Oral Questioning (OQ)

Five assessor-led questions mapped to K1–K4, with follow-up probes where needed.

Competent / Not Yet Competent • Assessor records evidence

Course Map



Eight labs turn the concepts into governed marketing and video campaign evidence.

Course Material Architecture

LMS-TMS PORTAL

Authenticated access point for the enrolled course and current resource links.

LEARNER PACKAGE

Visual slide PDF, Learner Guide and Lesson Plan published as versioned artifacts.

HANDS-ON WORKSPACE

Eight self-contained lab folders with mock data, procedures, verification and cleanup.

INTEGRITY SIGNALS

Current version, complete inventory, redacted evidence and no secrets or trainer answer keys.

Learner Repository

Open the repository for all eight lab folders

<https://github.com/tertiarycourses/TGS-2020503395-Business-Innovation-with-AI-Agents>

The repository contains learner materials only. Assessment answers, credentials and private evidence are excluded.

Classroom Setup Boundary

1

Use fictional data

No customer records, personal data or live production content.

2

Keep keys outside labs

Use .env.example templates; never submit .env or screenshots of secrets.

3

Start constrained

Enable only the toolsets required by the current lab.

4

Stop on mismatch

A wrong course, folder, account or destination is a blocker—not a reason to guess.

DAY 1

AI agents as governed business systems

Compare the engineering eras, identify opportunity, configure Hermes, orchestrate specialists and retain human authority.

AI Engineering Evolution and Agent Architecture

PROMPT → CONTEXT → HARNESS → AGENT

Trace the engineering shift from single responses to governed goal-directed systems and map the human brain-body analogy to model-harness design.

WEIGHT 20% · LABS 1 / 2

2023–2026 engineering timeline



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Prompt engineering and context engineering

Prompt engineering

- ✓ Instruction wording
- ✓ Examples and output schema
- ✓ Single-turn optimisation
- ✓ Failure: underspecified intent

Context engineering

- ✓ Evidence selection
- ✓ Memory and retrieval
- ✓ Tool results and state
- ✓ Failure: irrelevant or stale context

Harness engineering and agent engineering

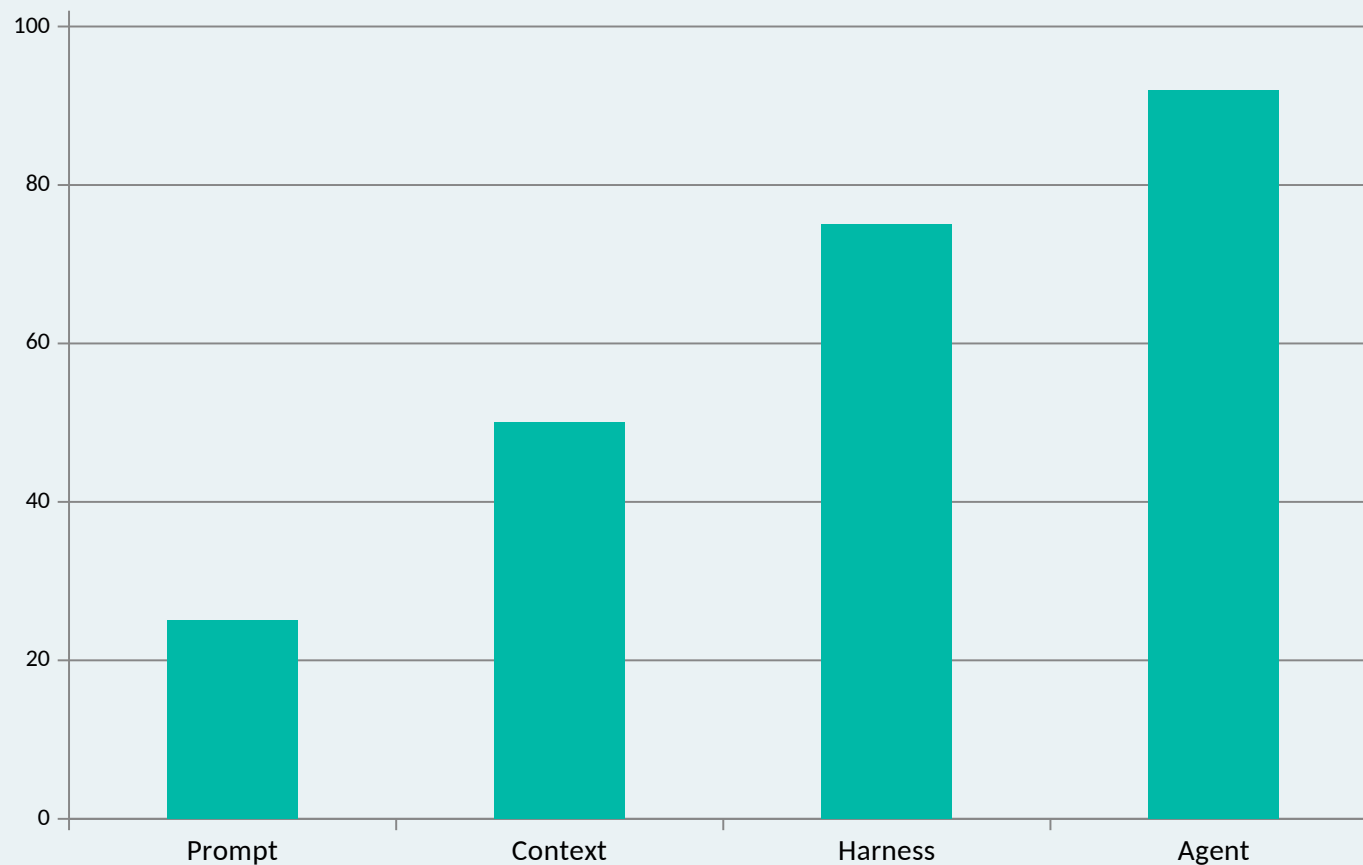
Harness engineering

- ✓ Tool registry and permissions
- ✓ Session lifecycle and retries
- ✓ Observability and approvals
- ✓ Deterministic runtime controls

Agent engineering

- ✓ Goal decomposition
- ✓ Dynamic tool selection
- ✓ Observation and replanning
- ✓ Human accountability for impact

Engineering leverage shifts outward

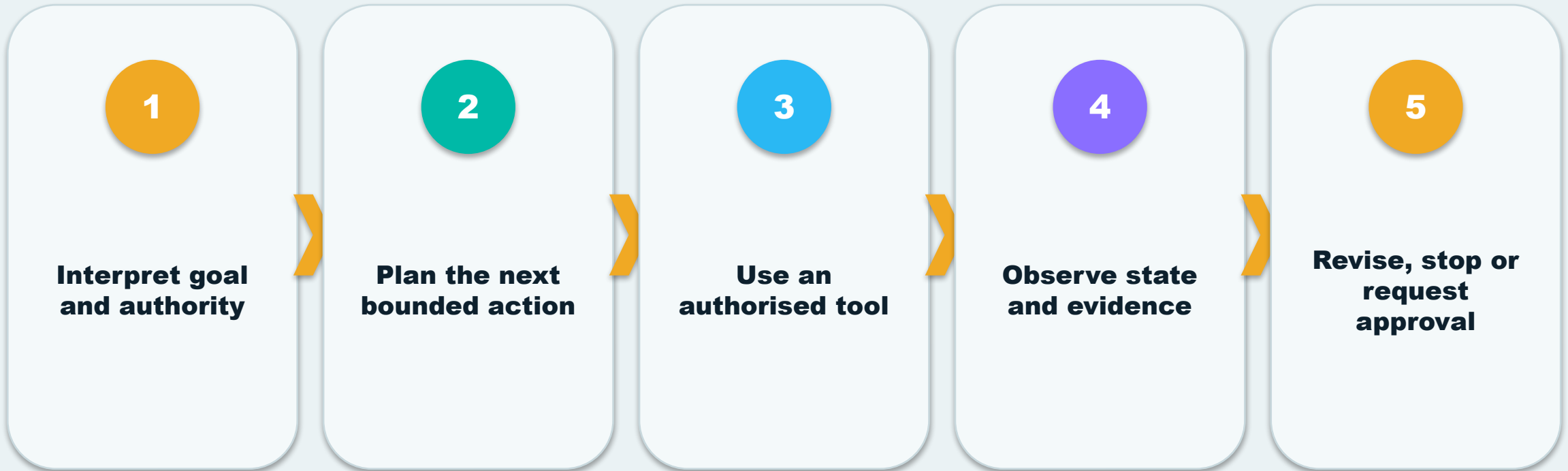


WHAT THE DATA SHOWS

Business reliability increasingly depends on the system around the model, not only the model response.

Illustrative teaching index — use the relationship, not the values, to guide design decisions.

Agent operating loop



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Human and AI-agent operating systems

Human: brain + body

- ✓ Brain reasons from perception and memory
- ✓ Body senses and acts in the world
- ✓ Values and accountability remain human
- ✓ Fatigue and attention constrain execution

AI agent: model + harness

- ✓ Model predicts and reasons over context
- ✓ Harness exposes tools, state and controls
- ✓ No independent moral or legal accountability
- ✓ Compute and data constrain execution

The harness has six control planes

1

Context

Instructions, files, retrieval and current state.

2

Tools

Typed actions for files, web, APIs and applications.

3

Memory

Curated facts and past evidence across sessions.

4

Orchestration

Planning, delegation, scheduling and handoffs.

5

Safety

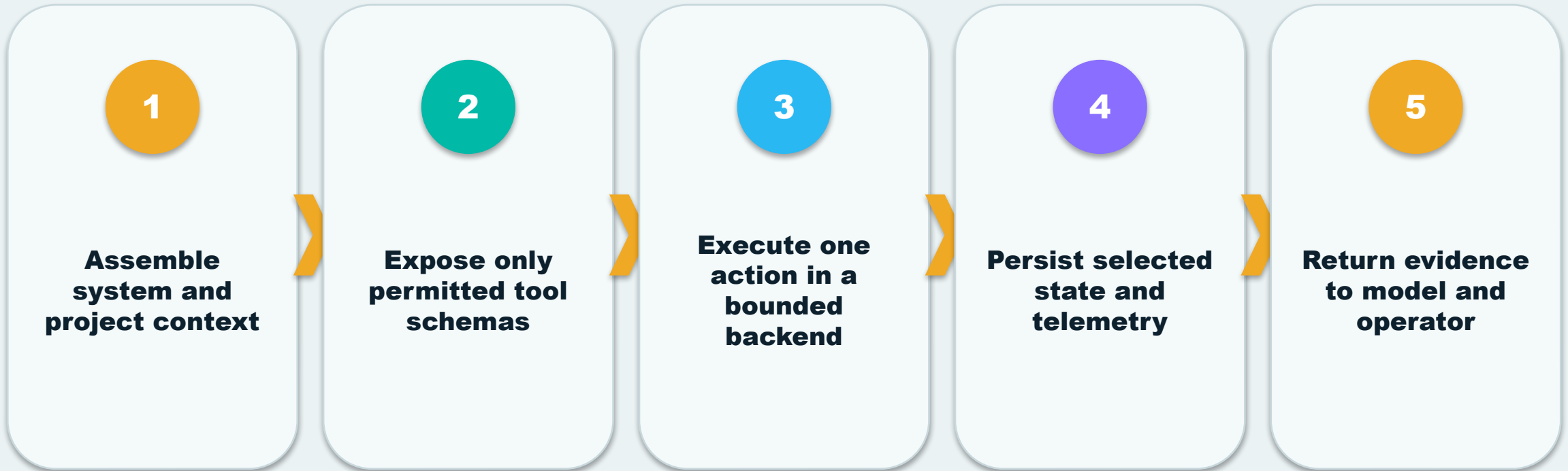
Permissions, sandboxing, approvals and policy.

6

Observability

Logs, traces, tests, cost and outcome metrics.

What the harness does on each turn



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Where agent systems fail

1

Goal

Ambiguous outcome or no stop condition.

2

Grounding

Missing, stale or poisoned evidence.

3

Authority

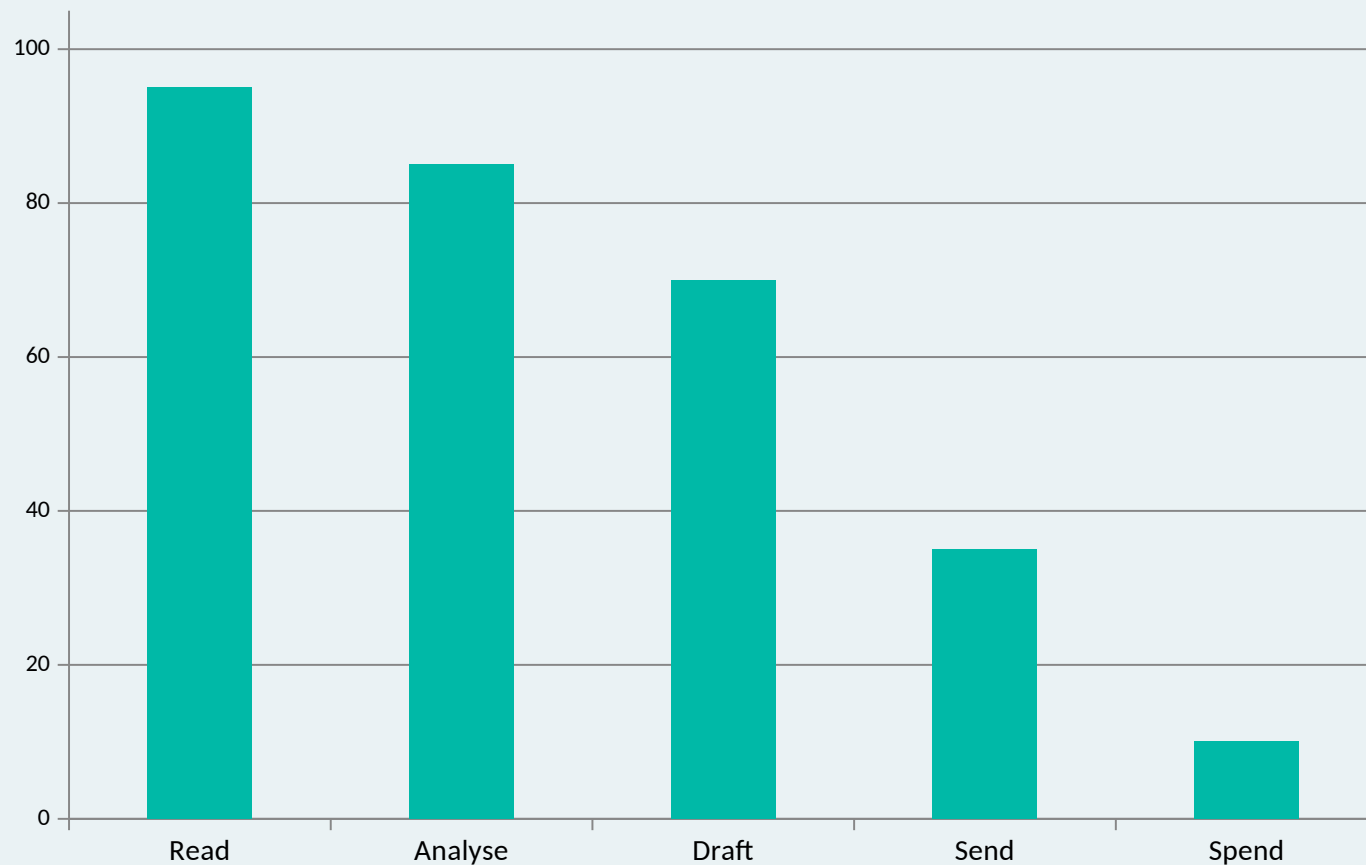
Tool scope exceeds business need.

4

Verification

Plausible narrative replaces observed state.

Autonomy follows consequence



WHAT THE DATA SHOWS

As reversibility falls and external impact rises, explicit human approval should increase.

Illustrative teaching index — use the relationship, not the values, to guide design decisions.

Four kinds of agent evidence

1

Input provenance

Where facts, files and instructions came from.

2

Execution trace

Which tool ran with which bounded arguments.

3

State proof

What file, record or status changed.

4

Decision record

Who approved, rejected or revised the outcome.

Model plus harness in the workplace



The model proposes; the harness constrains, acts, records and returns control to people.

- 1 Editable labels stay in PowerPoint
- 2 Approval is a system state
- 3 Evidence closes the loop

Copilot and agent

Copilot interaction

- ✓ Human initiates each turn
- ✓ Recommendation is the main output
- ✓ Tools are often user-triggered
- ✓ Context usually ends with the session

Agent operation

- ✓ Goal can span several actions
- ✓ State changes can be the output
- ✓ Tools are selected during the loop
- ✓ Sessions, memory and schedules create continuity

Safe task contract



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Compare Prompt, Context and Harness Controls

OBJECTIVE

Turn an ambiguous request into a measurable task contract and a minimal context package for an agent.

DELIVERABLE

A prompt-context comparison, AGENTS.md context file, acceptance rubric and a redacted test transcript.

TOOLS

Hermes · context files · session evidence

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

The improved run names its evidence, respects the no-send rule and passes every acceptance item; the learner distinguishes prompt wording from context and runtime authority.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

Explain one failure, one control and one improvement.



Handoff

Another learner can understand what was built and how success was verified.

Install Hermes Agent and Run a Safe First Session

OBJECTIVE

Install Hermes through the official path, complete provider setup, verify the runtime and run a bounded read-only task.

DELIVERABLE

An installation verification record with version, doctor result, provider route, session evidence and cleanup status.

TOOLS

Hermes CLI • provider • local workspace

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

Hermes starts, diagnostics identify a configured provider, the first response uses only fictional evidence, and installation-verification.md is complete.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

Explain one failure, one control and one improvement.



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Another learner can understand what was built and how success was verified.

10 technical workflow blueprints

Each blueprint exposes ingress, runtime capabilities, state/evidence and the control that stops unsafe side effects.

01 **Customer and Revenue**

02 **Operations and Supply**

03 **Regulated Knowledge Work**

04 **People and Public Service**

05 **Marketing and Media**

Runtime Surfaces Across the Reference Implementations

Blueprints using surface



ENGINEERING READING

Capabilities compose into a runtime graph.

- File and web tools connect reasoning to evidence.
- Delegation divides work into bounded specialist roles.
- Memory adds continuity and a new governance boundary.

Customer and Revenue

01

Retail service recovery

MEDIUM

INGRESS

New complaint with order ID

CAPABILITY
GRAPH

CRM MCP · policy skill · memory

STATE /
EVIDENCEOrder source, policy citations,
approver and final action

CONTROL GATE

Agent cannot refund or send
without approval

02

B2B lead intelligence

MEDIUM

INGRESS

Approved target-account list

CAPABILITY
GRAPH

Web · delegation · CRM MCP

STATE /
EVIDENCESource URLs, scoring rationale
and approval state

CONTROL GATE

Sales owner approves contact and
claims

Operations and Supply

03

Manufacturing exception triage

HIGH

INGRESS

Quality or downtime alert

CAPABILITY
GRAPH

Files · data MCP · skill

STATE /
EVIDENCESignal IDs, thresholds, SOP version
and decision

CONTROL GATE

Engineer authorises any
equipment action

04

Logistics disruption planner

HIGH

INGRESS

Late shipment or route closure

CAPABILITY
GRAPH

Web · logistics MCP · calculator

STATE /
EVIDENCEShipment IDs, rates, assumptions
and approval

CONTROL GATE

Operations owner approves
reroute and spend

Regulated Knowledge Work

05

Banking KYC review pack

HIGH

INGRESS

Incomplete customer due-diligence file

CAPABILITY
GRAPH

Document tools · rules skill · audit log

STATE /
EVIDENCE

Document list, rule references and reviewer decision

CONTROL GATE

Compliance officer makes disposition

06

Healthcare discharge education

HIGH

INGRESS

Approved discharge summary

CAPABILITY
GRAPH

Files · approved knowledge · language skill

STATE /
EVIDENCE

Source summary, version, clinical approval and delivery

CONTROL GATE

Clinician validates accuracy and suitability

People and Public Service

07

HR policy navigator

MEDIUM

INGRESS

Employee policy question

CAPABILITY
GRAPH

RAG · files · memory

STATE /
EVIDENCEPolicy version, citations,
confidence and escalation

CONTROL GATE

HR decides eligibility and
exceptions

08

Municipal case intake

MEDIUM

INGRESS

Resident service request

CAPABILITY
GRAPH

Forms · geodata MCP · workflow

STATE /
EVIDENCECase ID, data basis and routing
decision

CONTROL GATE

Officer confirms category and
priority

Marketing and Media

09

Multi-agent marketing campaign

MEDIUM

INGRESS

Approved product and audience brief

CAPABILITY
GRAPH

Delegation • web • skills • files

STATE /
EVIDENCE

Sources, claim register, revisions and sign-off

CONTROL GATE

Brand owner approves claims, spend and release

10

Multi-agent video campaign

HIGH

INGRESS

Approved campaign concept

CAPABILITY
GRAPH

Delegation • image/video tools • files

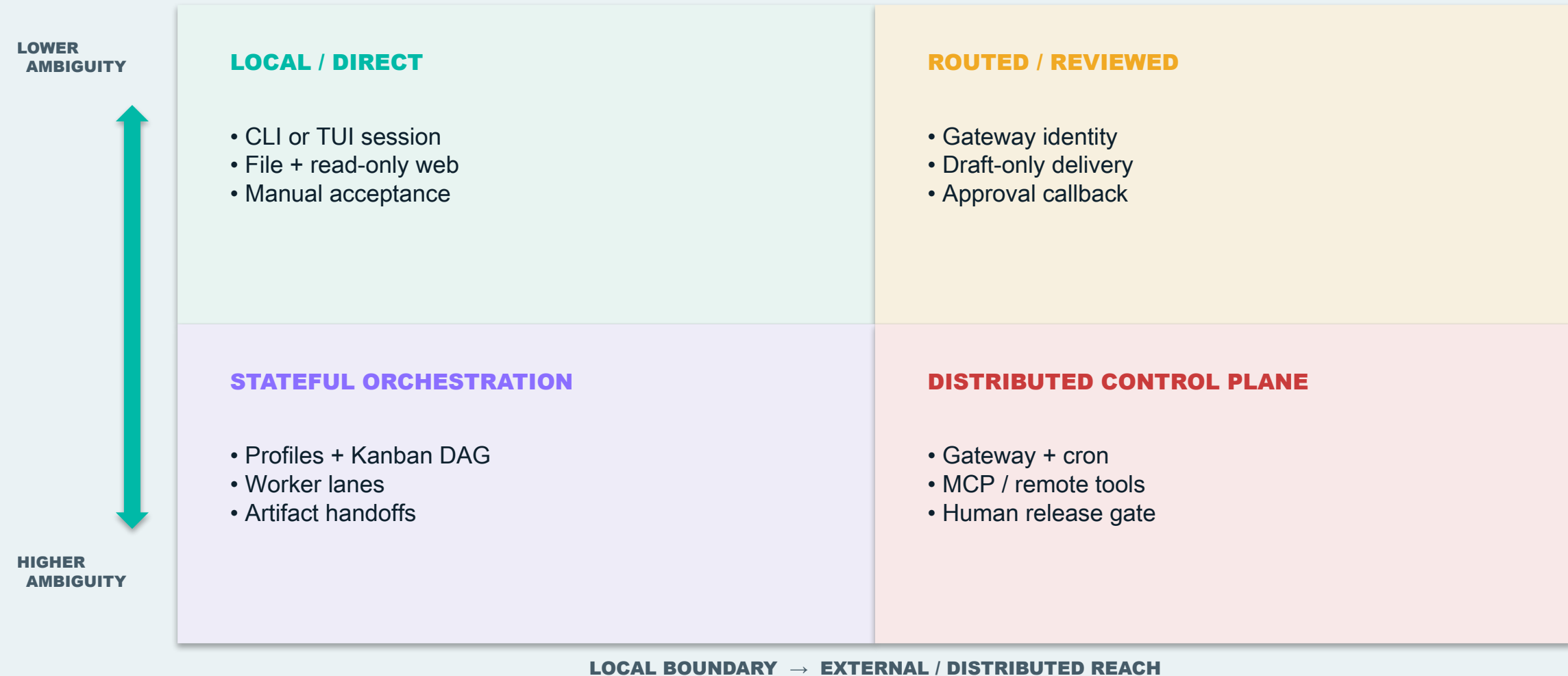
STATE /
EVIDENCE

Consent, licences, prompts, QC and release record

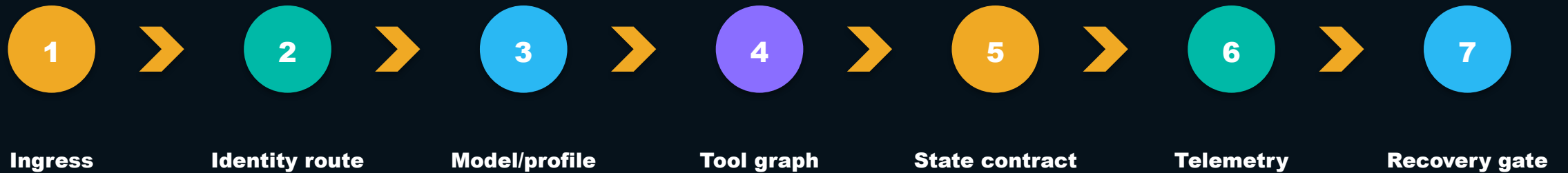
CONTROL GATE

Human approves likeness, rights and final render

Deployment Topology Selector



Seven interfaces make a Hermes workflow operable



Every interface must identify its owner, data contract, failure signal and observable proof.

Business Opportunities and Feasibility

PROCESS EVIDENCE · VALUE · COST · PILOT

Select realistic AI-agent opportunities from process evidence and build a risk-adjusted business case instead of automating attractive anecdotes.

WEIGHT 20% · LABS 6

Opportunity discovery funnel



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Agent-worthy task signals

1

Variable inputs

Cases differ enough to require interpretation.

2

Tool crossing

Work spans documents, systems or channels.

3

Feedback

Intermediate results change the next action.

4

Evidence

The result can be tested against observable criteria.

Use automation when the path is known

Conventional automation

- ✓ Stable rules
- ✓ Structured inputs
- ✓ Low ambiguity
- ✓ High repeat volume

AI agent

- ✓ Changing evidence
- ✓ Unstructured inputs
- ✓ Several valid paths
- ✓ Human review of judgement

Process evidence sheet

1

Baseline

Cycle time, volume, cost and error rate.

2

Bottleneck

Wait, rework, handoff or search delay.

3

Decision

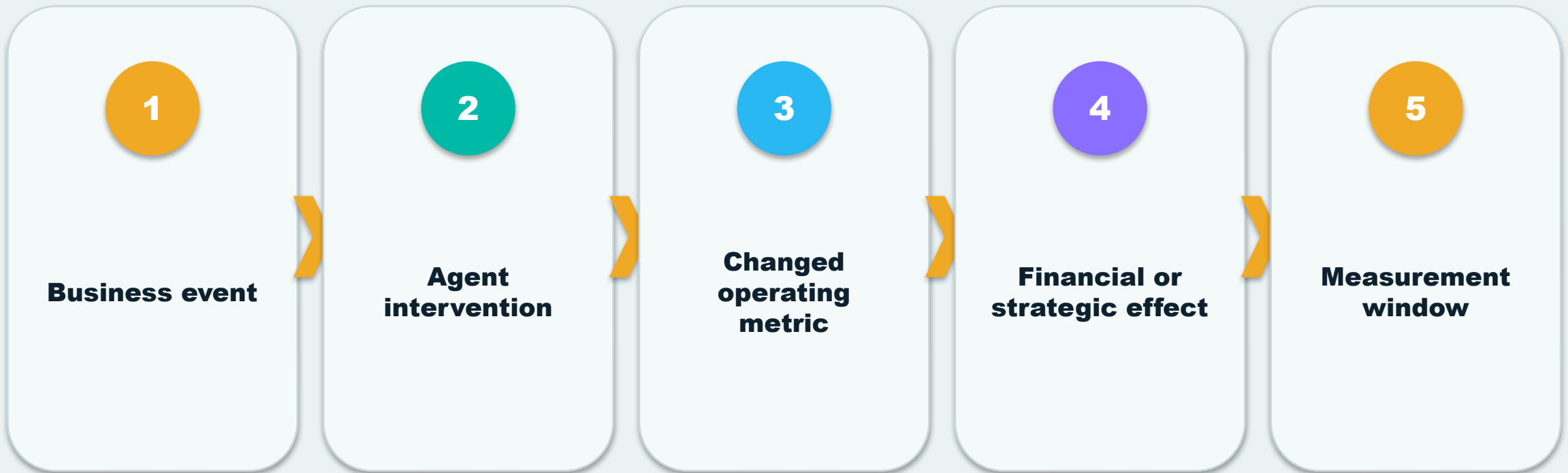
Who exercises judgement and with what policy.

4

Target

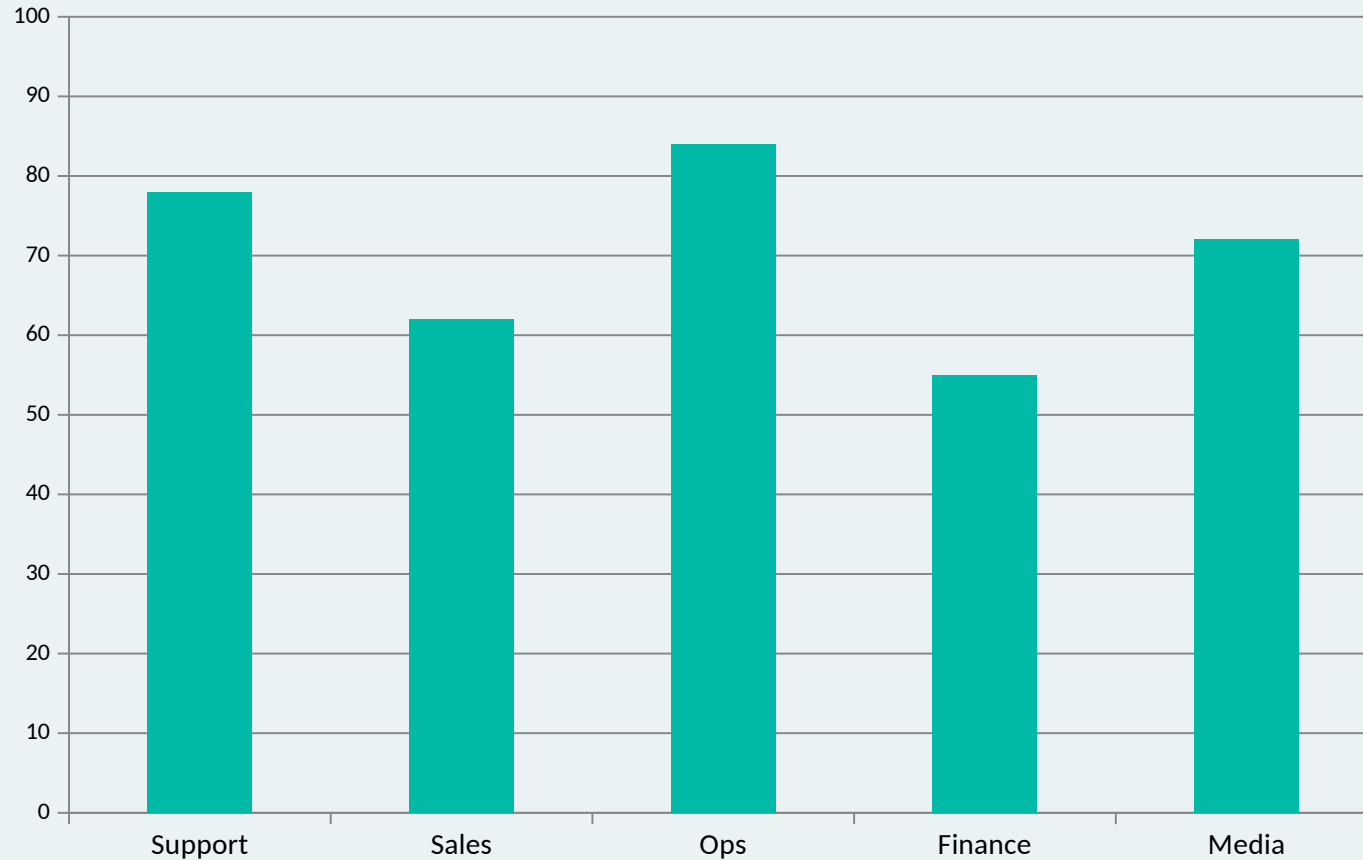
Specific measurable future-state change.

Value hypothesis



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Illustrative opportunity portfolio



WHAT THE DATA SHOWS

Prioritisation combines value, feasibility and governability; the tallest business value alone does not win.

Illustrative teaching index — use the relationship, not the values, to guide design decisions.

Feasibility has four lenses

1

Data

Availability, quality, rights and freshness.

2

Technical

Models, tools, integration and reliability.

3

Operational

Owners, exceptions, training and support.

4

Governance

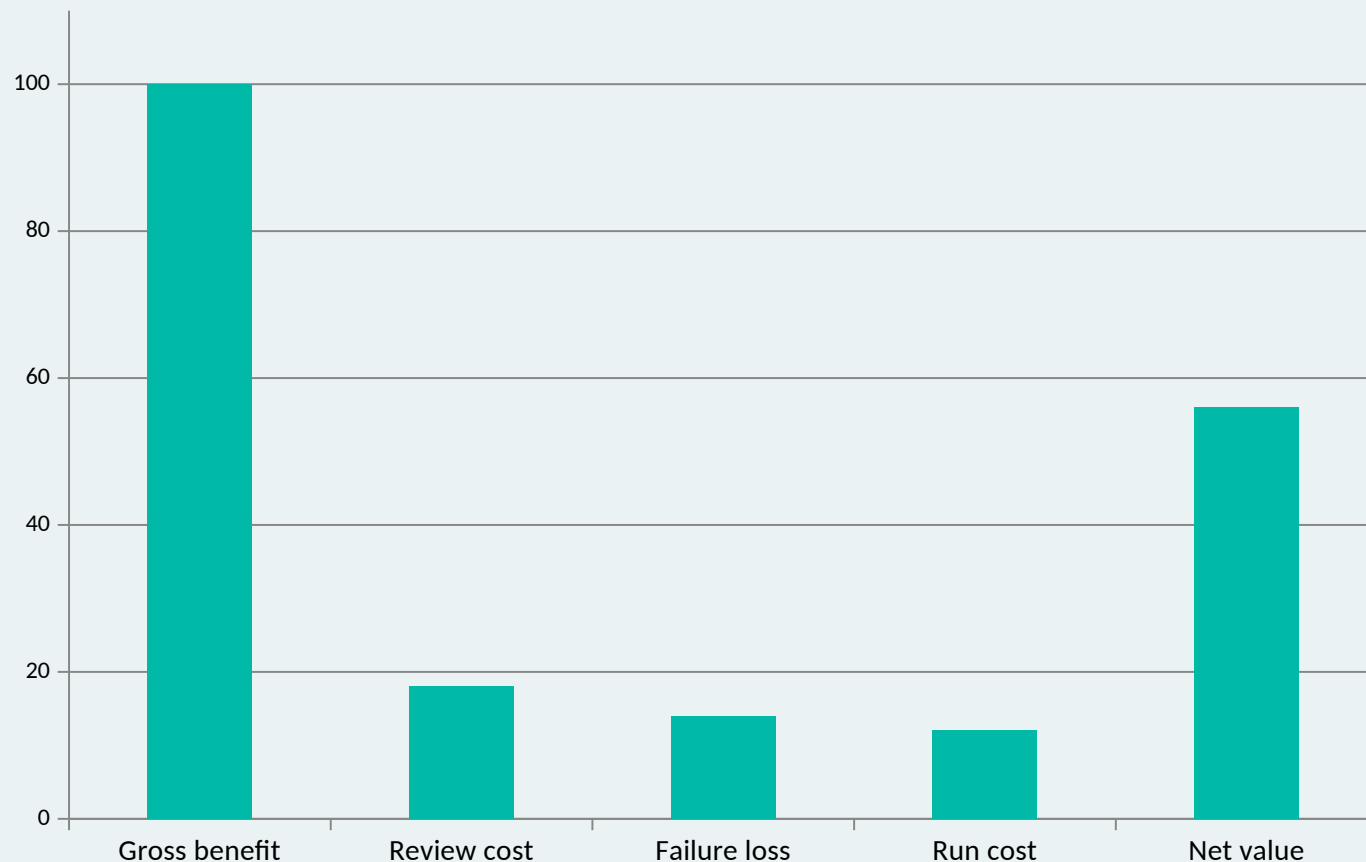
Privacy, security, compliance and accountability.

Pilot cost model



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Risk-adjusted benefit



WHAT THE DATA SHOWS

A credible business case subtracts review, expected failure loss and recurring operation—not just software fees.

Illustrative teaching index — use the relationship, not the values, to guide design decisions.

Expected value fields

1

Benefit

Hours saved × loaded cost, revenue lift or loss avoided.

2

Probability

Observed pilot success rate, not optimism.

3

Exposure

Volume and consequence per decision.

4

Control cost

Review, monitoring, incident and audit effort.

Proof of concept and pilot

Proof of concept

- ✓ Can the mechanism work?
- ✓ Small synthetic dataset
- ✓ Technical learning
- ✓ No business-scale claim

Pilot

- ✓ Does it create value in context?
- ✓ Representative bounded cases
- ✓ Operational and governance evidence
- ✓ Go, revise or stop decision

Pilot scorecard



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Prioritise Ten Industry Use Cases

OBJECTIVE

Score the ten realistic use cases for value, feasibility, risk and evidence readiness, then calculate a risk-adjusted pilot case.

DELIVERABLE

A completed opportunity matrix, transparent cost-benefit calculation, pilot charter and stop decision.

TOOLS

**Spreadsheet model ·
Hermes analysis · human
decision**

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

The selected case wins under the documented weights, the arithmetic is reproducible, risks and review cost are included, and the pilot has a measurable stop condition.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

Explain one failure, one control and one improvement.



Handoff

Another learner can understand what was built and how success was verified.

Ten-industry use-case test

1

Specific

Trigger, inputs and output are named.

2

Controllable

Human gate and tool scope are explicit.

3

Observable

Evidence shows whether work succeeded.

4

Economic

Value exceeds build, run and risk cost.

Scale or stop

Scale signals

- ✓ Repeatable quality
- ✓ Positive risk-adjusted value
- ✓ Low exception burden
- ✓ Named operating owner

Stop or redesign signals

- ✓ Unverifiable output
- ✓ Sensitive data without controls
- ✓ Review cost erases value
- ✓ Failures are hard to detect

Hermes Agent Harness in Practice

INSTALL · CONTEXT · TOOLS · SKILLS · MEMORY

**Inspect Hermes Agent as a concrete harness
and connect each runtime component to a
business control, learner lab and observable
output.**

WEIGHT 20% · LABS 1 / 2 / 3 / 4

Hermes Agent runtime surface

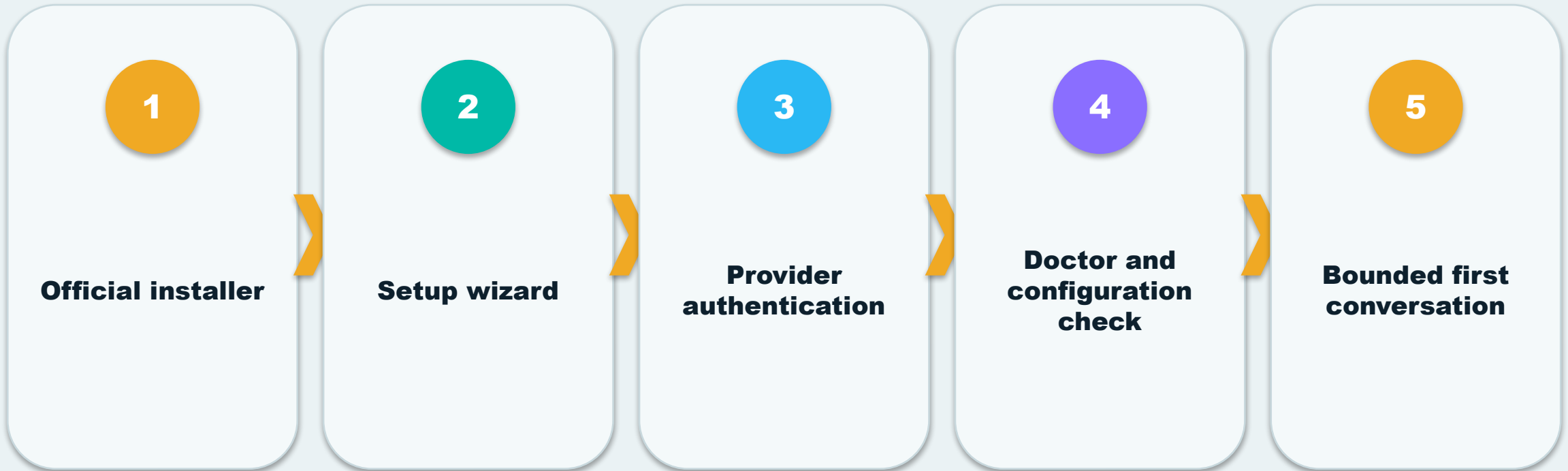


HERMES-AGENT

Hermes combines conversation, tool use, memory, skills and delegation inside one open-source harness.

- 1 Model-flexible
- 2 Local state by default
- 3 Toolsets are configurable

Supported installation path



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Install Hermes Agent and Run a Safe First Session

OBJECTIVE

Install Hermes through the official path, complete provider setup, verify the runtime and run a bounded read-only task.

DELIVERABLE

An installation verification record with version, doctor result, provider route, session evidence and cleanup status.

TOOLS

Hermes CLI · provider · local workspace

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

Hermes starts, diagnostics identify a configured provider, the first response uses only fictional evidence, and installation-verification.md is complete.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

Explain one failure, one control and one improvement.



Handoff

Another learner can understand what was built and how success was verified.

Hermes workspace ownership

1

Configuration

Provider, toolset and profile settings.

2

Sessions

Conversation and tool traces for retrieval.

3

Memory

Curated cross-session knowledge.

4

Skills

Reusable procedural instructions and assets.

Context and memory

Context

- ✓ Available now
- ✓ Project rules and task files
- ✓ Tool results and session tail
- ✓ May be compressed as the session grows

Memory

- ✓ Selected for later
- ✓ Preferences and durable facts
- ✓ Curated rather than full transcript
- ✓ Must avoid secrets and transient noise

Toolsets are capability boundaries

1

File

Read, write and patch within allowed roots.

2

Terminal

Run commands in a selected backend.

3

Web and browser

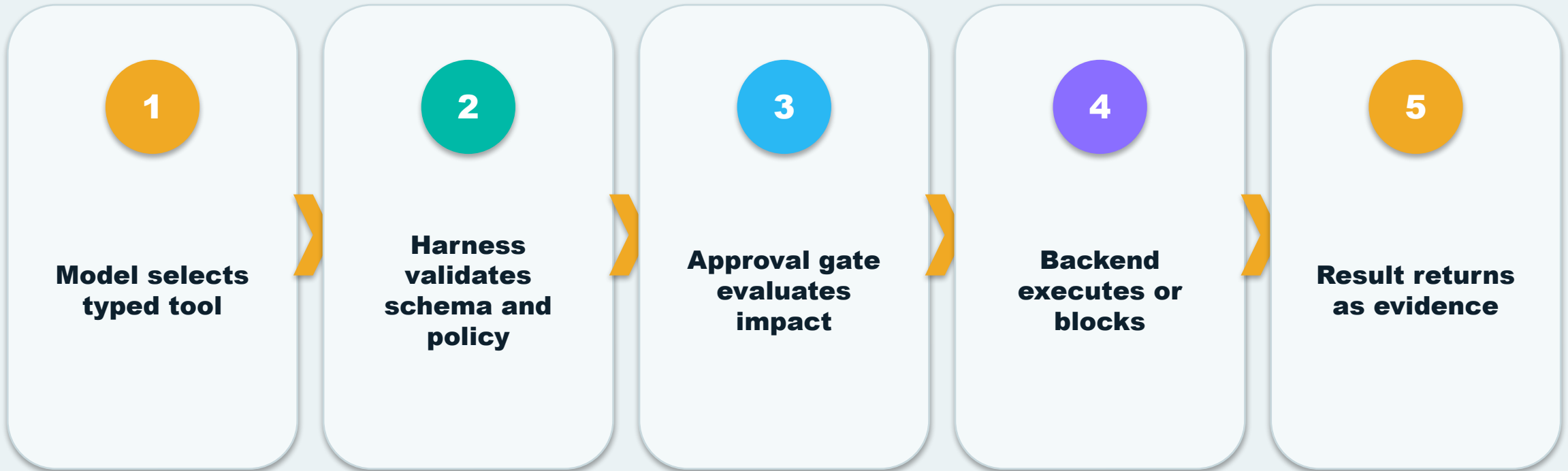
Retrieve changing evidence and operate pages.

4

Delegation

Spawn isolated specialist contexts.

Tool call lifecycle



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Demonstrate Tools and an MCP Design Boundary

OBJECTIVE

Use file and terminal tools for a grounded calculation, then design a least-privilege MCP connection without live credentials.

DELIVERABLE

A calculated service analysis, tool trace, MCP trust register and failure record.

TOOLS

Hermes file • terminal • MCP design

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

Independent calculations match the reported metrics, every claim points to the mock file, and the MCP design exposes only necessary capabilities.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

Explain one failure, one control and one improvement.



Handoff

Another learner can understand what was built and how success was verified.

Built-in tools and MCP

Built-in tool

- ✓ Ships with Hermes
- ✓ Known runtime semantics
- ✓ Platform availability checks
- ✓ Updated with Hermes

MCP tool

- ✓ External server boundary
- ✓ Server-defined schema
- ✓ OAuth or secret scope
- ✓ Trust and filtering need review

A useful skill contract

1

Trigger

When the procedure should load.

2

Scope

What the skill may read or change.

3

Method

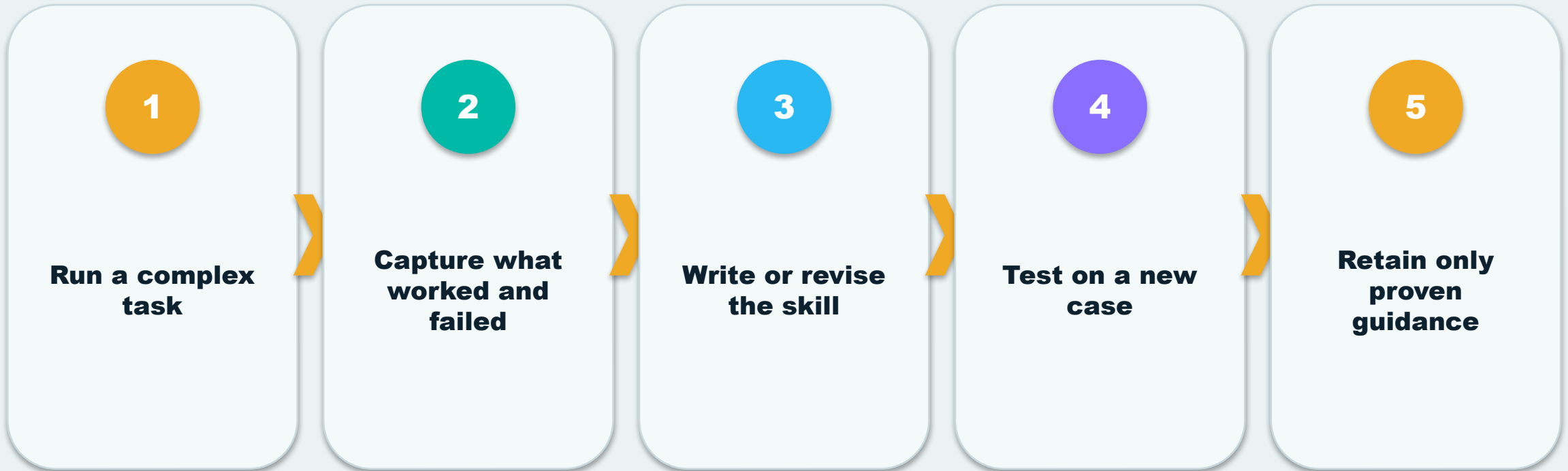
Reusable steps, templates and scripts.

4

Verification

Evidence and stop conditions.

Skill improvement loop



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Memory quality rules

1

Useful

Durable information that changes future work.

2

Attributable

Source and date are known.

3

Minimal

No secret, noise or redundant transcript.

4

Correctable

Review and update when evidence changes.

Demonstrate Memory Skills and Context Reuse

OBJECTIVE

Separate durable business memory from reusable procedural skills and verify both on a new fictional case.

DELIVERABLE

A memory decision log, a review-ready campaign skill, a second-case test and a correction record.

TOOLS

Hermes memory • skills • context files

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

The second case demonstrates durable preference reuse, procedural skill reuse, correction and zero secret or transient-noise retention.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

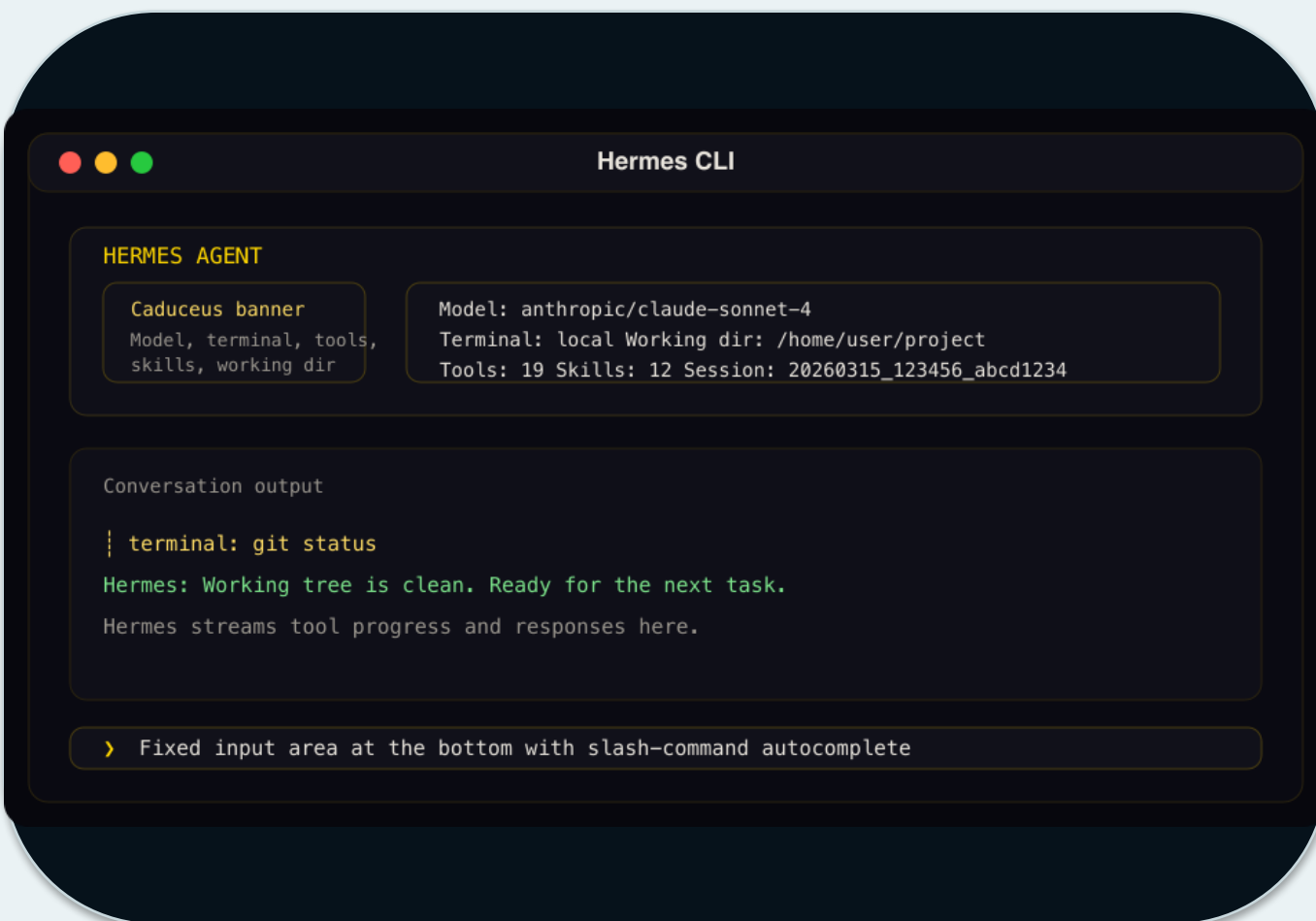
Explain one failure, one control and one improvement.



Handoff

Another learner can understand what was built and how success was verified.

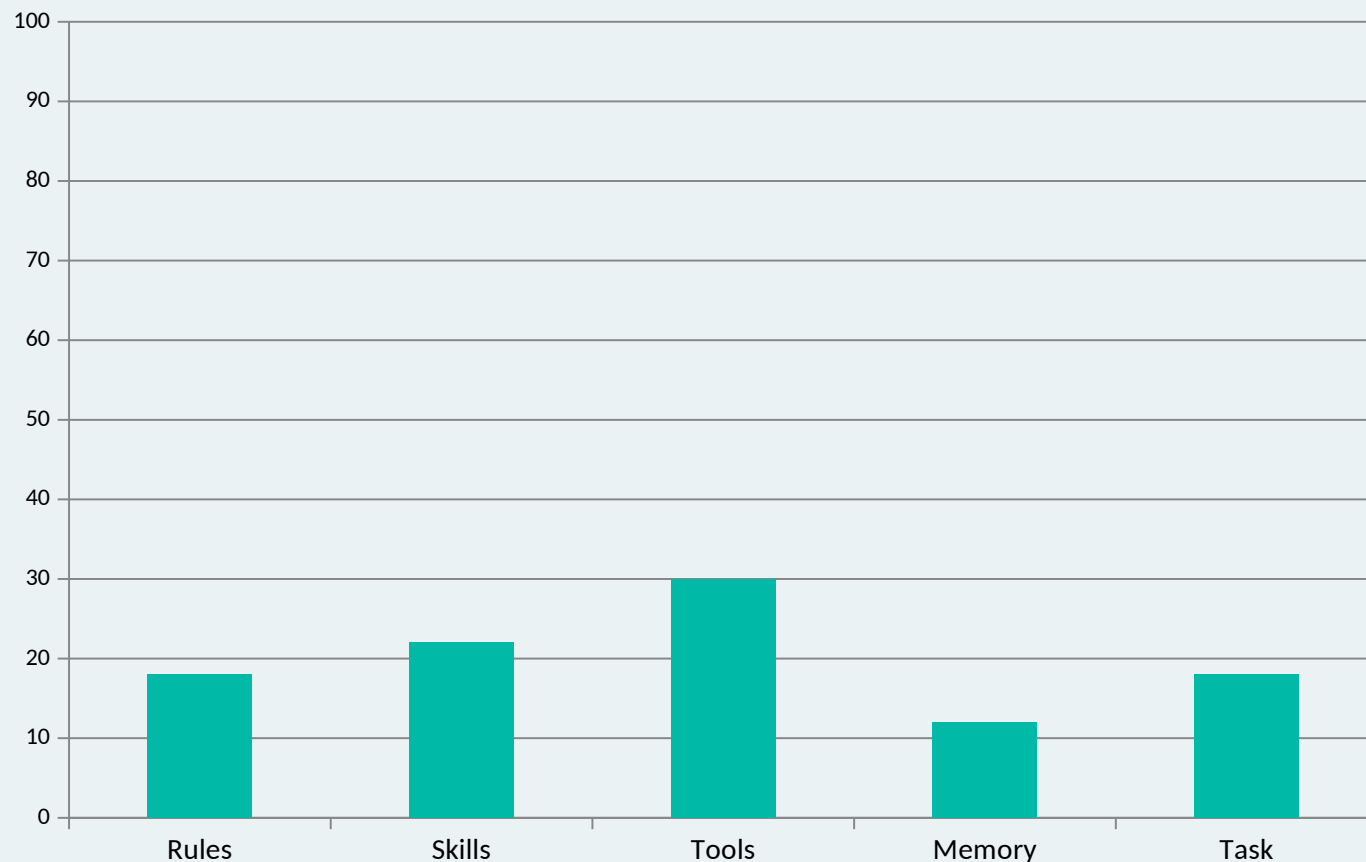
Hermes CLI as an operating cockpit



The CLI exposes conversation, tool evidence, approvals, usage and subagent state in one operational view.

- 1 Interruptible work
- 2 Visible tool results
- 3 Session continuity

Fixed context consumes the prompt budget



WHAT THE DATA SHOWS

Unused tool schemas and verbose instructions reduce the space available for task evidence.

Illustrative teaching index — use the relationship, not the values, to guide design decisions.

Grounded business answer



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Session resume and fresh session

Resume

- ✓ Preserves working history
- ✓ Good for interrupted work
- ✓ May carry stale assumptions
- ✓ Higher context cost

Fresh session

- ✓ Clean task boundary
- ✓ Good for scheduled or delegated work
- ✓ Needs explicit context package
- ✓ Easier evaluation

Compare Prompt, Context and Harness Controls

OBJECTIVE

Turn an ambiguous request into a measurable task contract and a minimal context package for an agent.

DELIVERABLE

A prompt-context comparison, AGENTS.md context file, acceptance rubric and a redacted test transcript.

TOOLS

Hermes · context files · session evidence

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

The improved run names its evidence, respects the no-send rule and passes every acceptance item; the learner distinguishes prompt wording from context and runtime authority.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

Explain one failure, one control and one improvement.



Handoff

Another learner can understand what was built and how success was verified.

Multi-Agent Campaign Orchestration

DELEGATE · CHALLENGE · SYNTHESISE · APPROVE

Design specialist agents for marketing and video campaigns with isolated context, structured handoffs and a human-controlled release boundary.

WEIGHT 20% · LABS 5 / 7 / 8

Delegation sequence



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Orchestrate Specialist Subagents

OBJECTIVE

Split one campaign problem into isolated research, audience and risk tasks with structured evidence handoffs.

DELIVERABLE

A task graph, three specialist handoffs, conflict-resolution note and human decision record.

TOOLS

**Hermes delegation •
profiles • Kanban**

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

Three role-bounded handoffs exist, conflicts remain visible, the synthesis waits for dependencies and the human decision record controls release.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

Explain one failure, one control and one improvement.



Handoff

Another learner can understand what was built and how success was verified.

Specialist role contract

1

Mission

One clear output and audience.

2

Inputs

Approved sources and context package.

3

Tools

Minimum capabilities for the role.

4

Handoff

Schema, evidence and failure status.

Parallel and sequential work

Parallel

- ✓ Independent research strands
- ✓ Lower elapsed time
- ✓ Needs normalised handoffs
- ✓ Conflicts resolved later

Sequential

- ✓ One output feeds the next
- ✓ Clear dependency chain
- ✓ Longer elapsed time
- ✓ Earlier errors can propagate

Marketing campaign agent team



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Multi-agent marketing command center



A campaign becomes an evidence-linked pipeline, not one oversized prompt.

- 1 Research before claims
- 2 Specialists challenge each other
- 3 Publication remains human-
authorised

Campaign handoff schema

1

Claim

Exact statement and supporting source.

2

Audience

Need, context and exclusion.

3

Asset

Channel, copy, dimensions and owner.

4

Decision

Approve, revise or reject with rationale.

Create a Marketing Campaign with Multiple AI Agents

OBJECTIVE

Coordinate research, audience, strategy, copy and review agents to produce a source-backed campaign pack that stops before publication.

DELIVERABLE

A campaign brief, evidence map, three channel variants, claim review, experiment plan and human approval record.

TOOLS

**Hermes delegation • web
• skills • files**

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

The campaign pack contains three differentiated channel drafts, a traceable evidence map, an experiment hypothesis, visible unresolved assumptions and human-controlled release.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

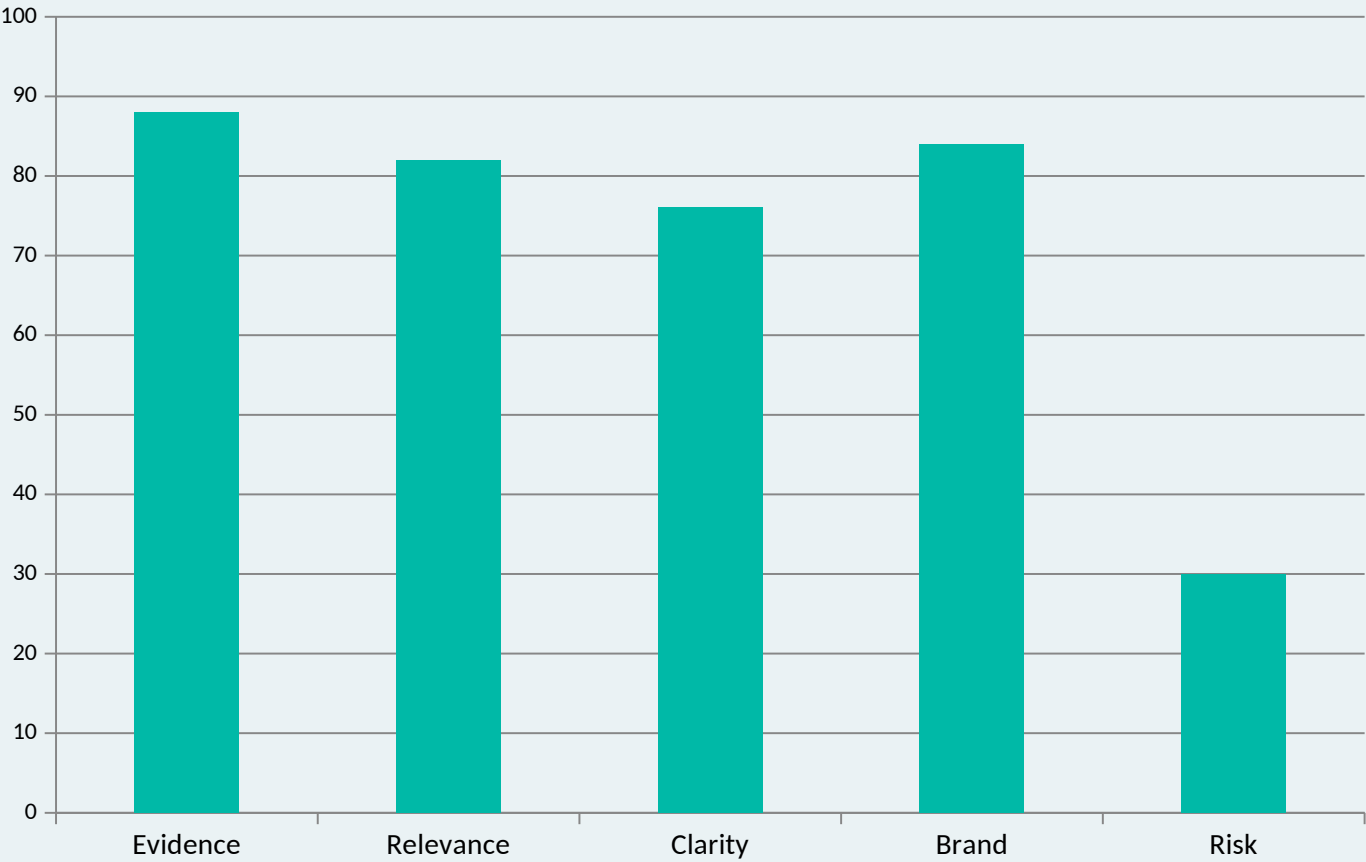
Explain one failure, one control and one improvement.



Handoff

Another learner can understand what was built and how success was verified.

Illustrative variant score



WHAT THE DATA SHOWS

Low risk score is desirable; the review gate balances performance with claim and brand exposure.

Illustrative teaching index — use the relationship, not the values, to guide design decisions.

Video campaign agent team



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Multi-agent video production studio



Video agents coordinate creative artifacts, but consent, rights and release remain human decisions.

- 1 Traceable asset manifest
- 2 Separate factual and visual QA
- 3 No unapproved likeness or music

Video production evidence

1

Storyboard

Scene objective, framing and continuity.

2

Asset register

Source, licence, consent and generation prompt.

3

QC report

Audio, captions, factual claims and brand checks.

4

Release record

Approved version, owner, date and destination.

Generated preview and approved deliverable

Preview

- ✓ Explores concept
- ✓ Watermarked or internal
- ✓ May use placeholders
- ✓ Not externally published

Deliverable

- ✓ Rights and consent cleared
- ✓ Claims verified
- ✓ Accessibility checked
- ✓ Named human approval

Conflict resolution between agents



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Orchestration failure modes

1

Context leak

Specialist receives unrelated sensitive data.

2

Silent dependency

Downstream work starts before required evidence.

3

Consensus illusion

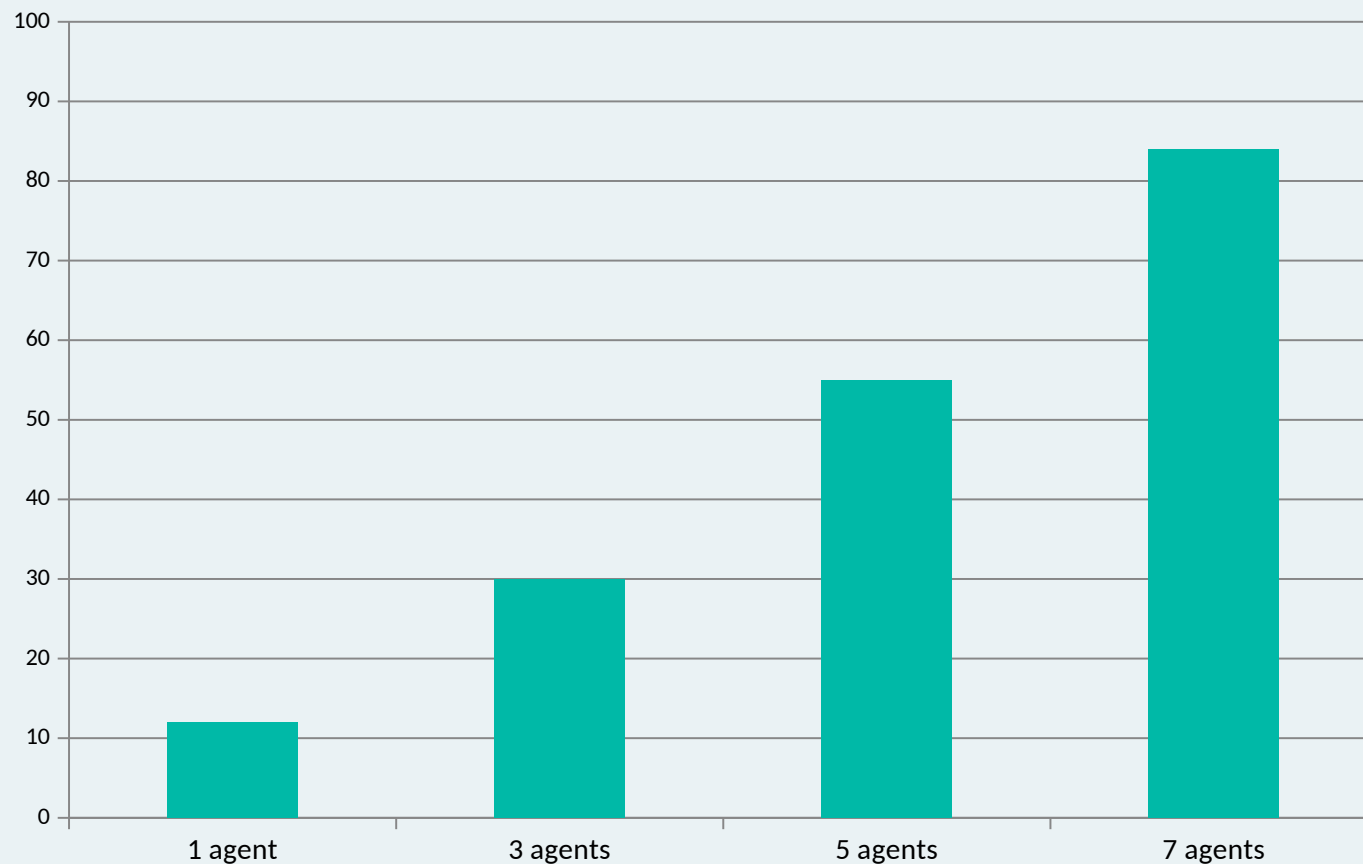
Several agents repeat the same unsupported claim.

4

Release drift

Draft moves outward without final approval.

Coordination cost grows with handoffs



WHAT THE DATA SHOWS

Add specialists only when role separation or parallelism creates more value than coordination overhead.

Illustrative teaching index — use the relationship, not the values, to guide design decisions.

Create a Governed Multi-Agent Video Campaign

OBJECTIVE

Produce a review-ready video campaign package with specialist agents, rights and consent controls, security tests and a human release gate.

DELIVERABLE

A script, storyboard, asset and rights register, generation plan, QC report, security matrix and final human release decision.

TOOLS

**Hermes delegation •
image/video tools •
security**

PROCEDURE

Detailed step-by-step instructions are in the Learner Guide and this lab folder.

Evidence and Acceptance Criteria



Working output

The production package is complete and traceable, no unapproved asset or claim is used, negative controls block unsafe actions, and only a named human can approve final release.



Evidence pack

Screenshots, generated files and completed checklist contain no secrets.



Reflection

Explain one failure, one control and one improvement.



Handoff

Another learner can understand what was built and how success was verified.

Security Governance and Implementation

THREATS · CONTROLS · OVERSIGHT · EVALUATION

Apply security, data governance, human oversight and continuous evaluation to turn a promising agent workflow into an accountable business process.

WEIGHT 20% · LABS 8

Agent attack surface

1

Instructions

Prompt injection and conflicting authority.

2

Data

Sensitive input, poisoned retrieval and leakage.

3

Tools

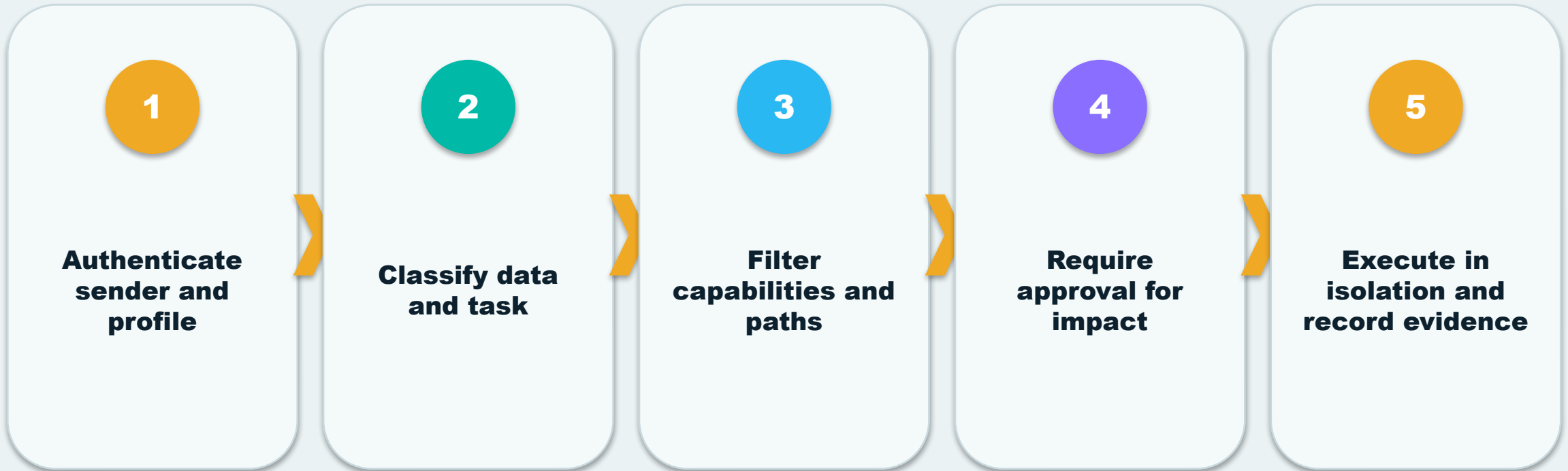
Over-privileged actions and unsafe arguments.

4

Supply chain

Untrusted skills, plugins and MCP servers.

Security enforcement order



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Model refusal and runtime control

Model refusal

- ✓ Probabilistic behaviour
- ✓ May vary by provider
- ✓ Useful defence in depth
- ✓ Not proof an action was blocked

Runtime control

- ✓ Deterministic policy check
- ✓ Observable allow or deny
- ✓ Applies before execution
- ✓ Produces auditable evidence

Least-privilege design

1

Identity

One profile and sender route per operating role.

2

Data

Only approved files, fields and retention.

3

Capability

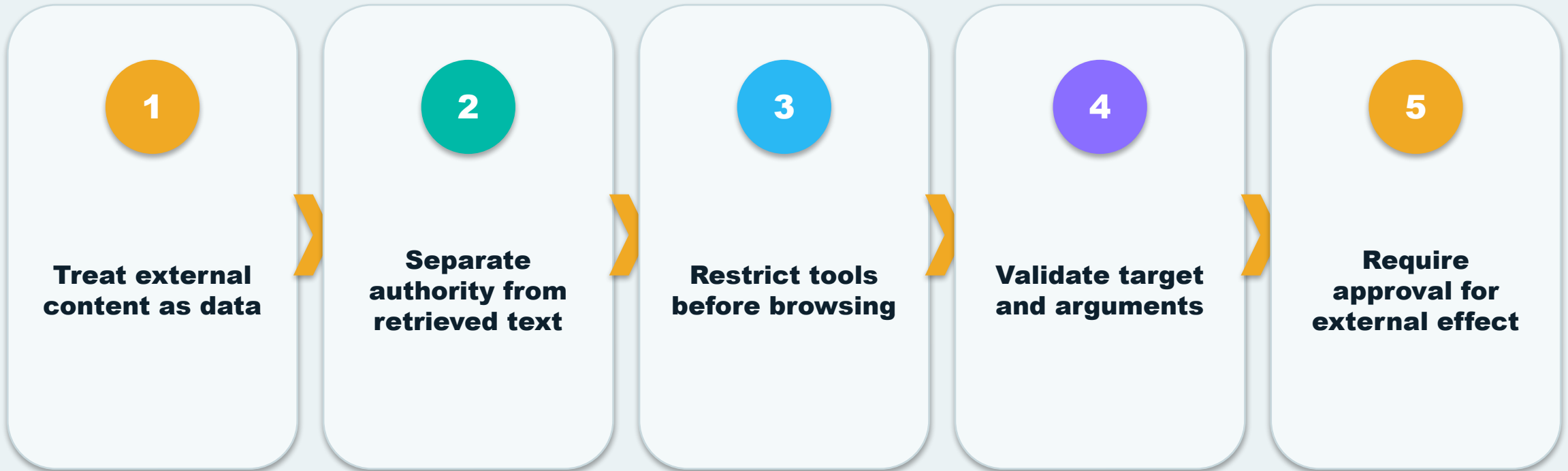
Only required tools and endpoints.

4

Time

Temporary approvals and credentials expire.

Prompt-injection defence



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Data governance lifecycle

1

Collect

Purpose, lawful basis and minimum fields.

2

Process

Approved provider, tools and locations.

3

Persist

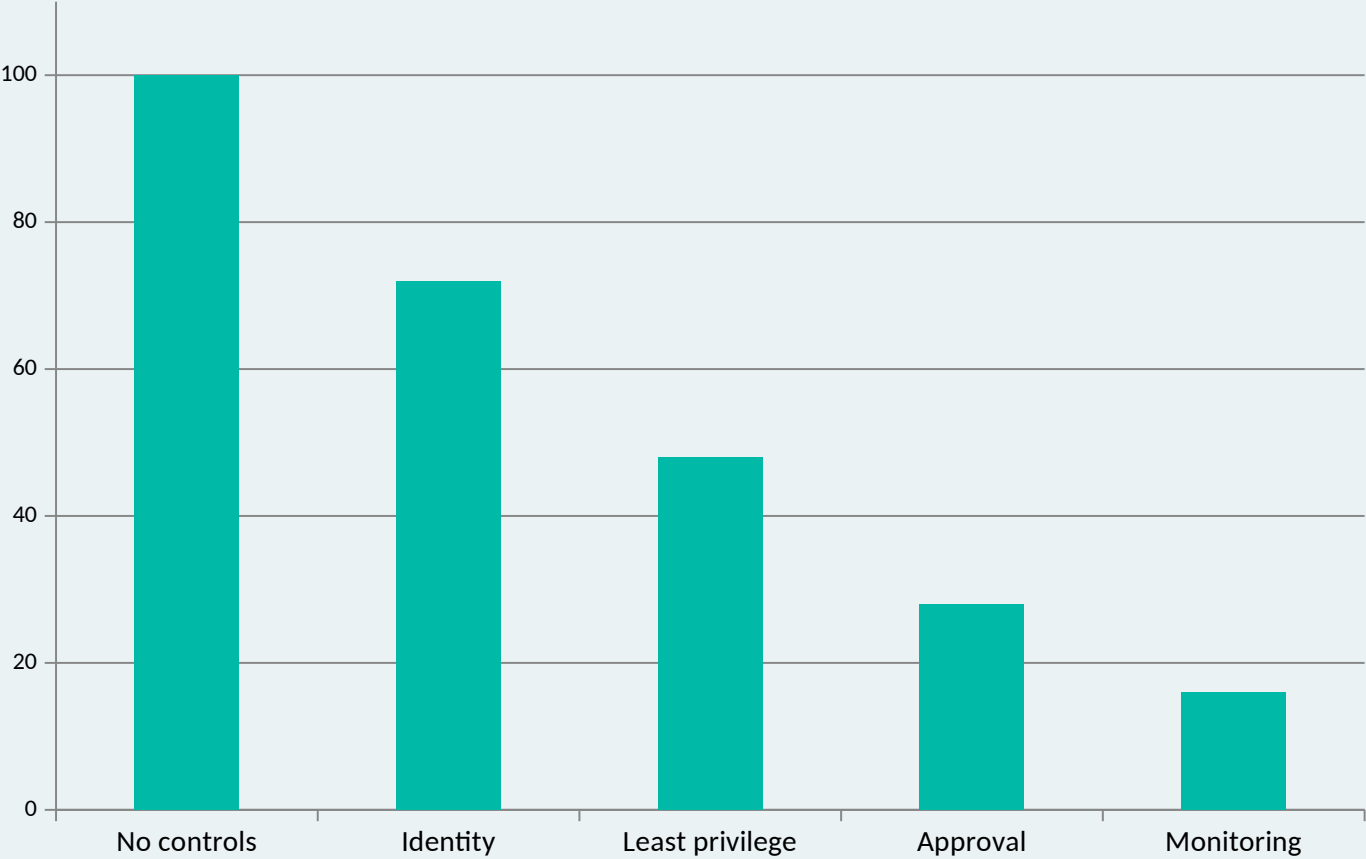
Session, memory, logs and access owners.

4

Delete

Retention trigger and verifiable removal.

Residual risk after layered controls



WHAT THE DATA SHOWS

Layered controls reduce exposure; monitoring is still required because no layer removes all risk.

Illustrative teaching index — use the relationship, not the values, to guide design decisions.

Human oversight roles

1

Requester

Defines need and permitted data.

2

Operator

Configures agent and watches execution.

3

Reviewer

Challenges evidence and output quality.

4

Accountable owner

Accepts residual risk and release decision.

AI incident response



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Monitor outputs and outcomes

Output metrics

- ✓ Factual accuracy
- ✓ Format completeness
- ✓ Tool success
- ✓ Latency and cost

Outcome metrics

- ✓ Cycle-time change
- ✓ Customer or user effect
- ✓ Review burden
- ✓ Risk events and reversals

Implementation workstreams

1

Process

Future-state design and exception handling.

2

People

Roles, training and change adoption.

3

Technology

Integration, evaluation and support.

4

Governance

Risk acceptance, audit and review cadence.

Pilot-to-scale decision



A useful agent does not stop at a plausible answer; it closes the loop with observable evidence.

Human body and agent harness under stress

Human recovery

- ✓ Pause and seek help
- ✓ Use judgement and values
- ✓ Learn from feedback
- ✓ Remain accountable

Agent recovery

- ✓ Stop condition and checkpoint
- ✓ Ask for clarification or approval
- ✓ Restore known state
- ✓ Return evidence to accountable human

Release gate evidence

1

Identity

Correct course, profile, owner and destination.

2

Integrity

Current artifact version and acceptance tests.

3

Privacy

No secrets, personal data or trainer-only content.

4

Preservation

Unrelated records and systems remain unchanged.

Create a Governed Multi-Agent Video Campaign

OBJECTIVE

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DELIVERABLE

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TOOLS

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Working output

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Evidence pack

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Reflection

Explain one failure, one control and one improvement.



Handoff

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AI-Agent Innovation Proof Chain



Innovation is complete only when the intended business outcome is supported by observable evidence, acceptable residual risk and named human approval.

FINAL ASSESSMENT

Demonstrate business judgement and knowledge



Written Assessment

30 MIN

Five questions • A1–A5



Oral Questioning

30 MIN

Five questions • K1–K4

Use the current assessor-issued papers. Do not share candidate responses or answer keys.

Assessment Flow



Complete the final digital attendance

The trainer will display the live TRAQOM assessment-attendance QR code. Scan that code, submit the record, then confirm receipt with your assessor.

LIVE QR

TRAINER DISPLAY

Do not scan this placeholder

THANK YOU

Build agents that earn trust

Clear outcome · minimum authority · grounded context · observable action · human accountability

<https://www.tertiarycourses.com.sg/wsqa-business-innovation-with-ai-agents.html>